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Random Survival Forests of Competing Risks

by

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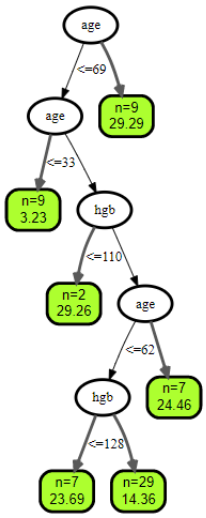

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Abstract

Competing risks occur in the financial, insurance, and especially, the medical field with often long-term observation periods. Initially, statistical models were developed to estimate and analyze the occurrence of an event when competing risks are present. With the increasing interest in machine learning (ML), ML methods were developed to analyze and predict competing risks events.

The goal of this thesis is to introduce basics of survival analysis that deal with competing risks. Then we introduce in detail a machine learning approach called Competing risks forests. Finally the thesis briefly introduces a statistical model approach called the cause-specific COX model to compare the two approaches. A part of the comparison is an application of both approaches to cancer data.

Competing risks forest is a nonparametric method which does not involve strict and unrealistic assumptions in the semiparametric COX model. Competing risks forest model is, however, less interpretable due to its "black box" nature compared to well-interpretable parameters in the COX model. Application of the two models to cancer data yields qualitatively similar results with almost identical conclusions, but quantitatively the estimated chances of event occurrence are quite different.

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Contents

1. Introduction **4**

2. Survival analysis of competing risks **6**
2.1. Competing risks data 6
2.2. Survival function, Cumulative incidence function, and Cause-specific hazard function 6
2.3. Nelson-Aalen, Kaplan-Meier, and Aalen-Johansen estimators 8
2.3.1. Nelson-Aalen estimator 9
2.3.2. Kaplan-Meier estimator 10
2.3.3. Aalen-Johansen estimator 11

3. Competing risks forests **12**
3.1. Growth of the competing risks tree 12
3.2. Splitting rule 13
3.2.1. Generalized log-rank test 13
3.2.2. Composite splitting rules 14
3.3. Construction of ensembles from a competing risks forest 14
3.4. Comparison to the COX model 15
3.5. Application to medical data 16

A. R-Code **21**

1. Introduction

Survival analysis is a statistical field developed to analyze and predict the time until an event occurs. This includes, among others, obtaining the probability of the occurrence of a certain event or identifying and comparing factors that influence this probability. The field comprises techniques to treat data in which each observation consists of a time until an event occurred, a type of an event that occurred, and additional characteristics of the observation. The strength of survival analysis rests on its versatility to handle censored data.

The techniques in survival analysis are tailored to the specifics of given data and their context. The following examples outline the variation in the models and techniques used in survival analysis as well as various settings where the usage of survival analysis is deemed to be beneficial.

- Statistics Norway preserves the year of marriage contraction and the possible subsequent divorce. Based on the data, Aalen et al. (2008) conclude that the proportions of still married decreased as the year of marriage contraction increased. This model considers only one possible event - divorce.

- Vollset et al. (2006) examined data that include 10-year smoking habits, sex, and the age and cause of death of 50 thousand Norwegian men and women. The study concludes that despite similar rates for lung cancer death, women who smoked had lower mortality rates in middle age than men with similar smoking histories due to fewer cardiovascular deaths in women. This model considers more competing events: death of cancer, cardiovascular death, and other deaths.

- In a study by Klein and Moeschberger (2003, section 3.2), 137 patients with acute leukemia received a bone marrow transplant. The data consist of the times to relapse and death, and the times when the platelet count of a patient returned to a self-sustaining level. This model uses Markov chains with three states: "transplanted", "platelet recovery", and "relapse or death". Every patient starts at the "transplanted" state and can move to either of the other states. A patient can also move from the "platelet recovery" state to the "relapse or death" state. Such model is called the illness-death model.

The survival analysis methods are developed to address challenges of survival data that come as a mixture of complete and incomplete observations. The data incompleteness arises from the real-world phenomena tied to long-time observation of events. Other statistical data may require only single-time measurements, e.g., measurement of the blood pressure of patients, which do not yield incomplete observations.

Survival data differ from other statistical data especially because of left-truncation and right-censoring. Left-truncated data arise when a study only includes subjects who have survived a minimum duration. For example, patients with myocardial infarction are only included in the study if they survive the initial phase and reach the hospital. Right-censored data can arise when individuals withdraw themselves from the study, are lost to follow-up, or the study closes. This thesis only progresses further with right-censored survival data of competing risks.

2. Survival analysis of competing risks

This chapter introduces right-censored competing risks data and related survival analysis variables. Then we derive or give intuition behind some of the functions and expressions used in the following chapters.

2.1. Competing risks data

Consider a study that records when each of the n observed subjects experiences one of multiple possible events. Each subject $i = 1, ..., n$ experiences an event type $\delta_i^o \in \{1, ..., J\}$ with $J \geqslant 1$ at some event time T_i^o. The events are mutually exclusive which means that once a subject experiences an event it cannot experience another event again. This thesis considers right-censored data as presented in Introduction. If the study has no record of the event that a subject experienced, it records the censoring time C_i^o which is the length of the time that the subject spends under observation before the subject is censored.

The competing risks data $(T_i, \delta_i, \mathbf{x}_i)_{1 \leqslant i \leqslant n}$ consist of time $T_i = \min(T_i^o, C_i^o)$, the event indicator $\delta_i = \delta_i^o I(T_i^o \leqslant C_i^o)$, and p-dimensional covariate vector $\mathbf{x}_i$ for each observed subject $i = 1, ..., n$. This means that either

- $T_i = C_i^o$ and $\delta_i = 0$ when the subject i is censored at time C_i^o, or
- $T_i = T_i^o$ and $\delta_i = j > 0$ when the study records that the subject i experienced the event j at time T_i^o.

The p-dimensional covariate vector $\mathbf{x}_i$ consists of p characteristics of the subject i.

We assume for the rest of this thesis that T^o and C^o are independent, and that T^o and C^o are continuously-distributed, which implies that $T = \min(C^o, T^o)$ is continuously-distributed. The independence means that the time when censoring occurs does not affect and is not affected by the time when one of the events occurs.

2.2. Survival function, Cumulative incidence function, and Cause-specific hazard function

The event-free survival function $S(t|\mathbf{x}) = P(T^o \geqslant t|\mathbf{x})$ is the probability that a subject with covariates $\mathbf{x}$ has not experienced any of the events up until time t. The cause-specific

cumulative incidence function (CIF) $F_j(t|\mathbf{x}) = P(T^o \leqslant t, \delta^o = j|\mathbf{x})$ is the probability that a subject with covariates $\mathbf{x}$ has experienced event j by time t. The CIFs complement the survival function, i.e.,

$$S(t|\mathbf{x}) + \sum_{j=1}^{J} F_j(t|\mathbf{x}) = 1 \tag{2.1}$$

holds for all times t and covariates $\mathbf{x}$.

The cause-specific hazard function

$$\alpha_j(t|\mathbf{x}) := \lim_{\Delta t \longrightarrow 0+} \frac{P(t \leqslant T^o \leqslant t + \Delta t, \delta^o = j|T^o \geqslant t, \mathbf{x})}{\Delta t} \tag{2.2}$$

is the measure of the instantaneous risk of event j occurring to a subject with covariates $\mathbf{x}$ at time t given that the event has not yet occurred. Multiplication of the nominator and denominator of Definition (2.2) by the survival function $S(t|\mathbf{x}) = P(T^o \geqslant t|\mathbf{x})$ yields

$$\begin{aligned}
\alpha_j(t|\mathbf{x}) &= \lim_{\Delta t \longrightarrow 0+} \frac{P(t \leqslant T^o \leqslant t + \Delta t, \delta^o = j|T^o \geqslant t, \mathbf{x})}{\Delta t} \frac{P(T^o \geqslant t|\mathbf{x})}{P(T^o \geqslant t|\mathbf{x})} \\
&= \lim_{\Delta t \longrightarrow 0+} \frac{P(t \leqslant T^o \leqslant t + \Delta t, \delta^o = j|\mathbf{x})}{P(T^o \geqslant t|\mathbf{x})\Delta t} \\
&= \frac{1}{P(T^o \geqslant t|\mathbf{x})} \lim_{\Delta t \longrightarrow 0+} \frac{P(T^o \leqslant t + \Delta t, \delta^o = j|\mathbf{x}) - P(T^o \leqslant t, \delta^o = j|\mathbf{x})}{\Delta t} \\
&= \frac{1}{S(t|\mathbf{x})} \lim_{\Delta t \longrightarrow 0+} \frac{F_j(t + \Delta t|\mathbf{x}) - F_j(t|\mathbf{x})}{\Delta t} \\
&= \frac{f_j(t|\mathbf{x})}{S(t|\mathbf{x})},
\end{aligned} \tag{2.3}$$

where $f_j(t|\mathbf{x}) := \frac{d}{dt} F_j(t|\mathbf{x})$ is the sub-distribution density for event j.

Fact. *For continuously-distributed event time T,*

$$F_j(t|\mathbf{x}) = \int_0^t S(s|\mathbf{x})\alpha_j(s|\mathbf{x})\mathrm{d}s = \int_0^t \exp\left(-\int_0^s \sum_{l=1}^{J} \alpha_l(u|\mathbf{x})\mathrm{d}u\right) \alpha_j(s|\mathbf{x})\mathrm{d}s. \tag{2.4}$$

Proof. Equation (2.3) gives $S(t|\mathbf{x})\alpha_j(t|\mathbf{x}) = f_j(t|\mathbf{x})$. Integrating both sides from 0 to t yields $\int_0^t S(s|\mathbf{x})\alpha_j(s|\mathbf{x})\mathrm{d}s = F_j(t|\mathbf{x})$, which proves the first equality in Fact (2.4).

By the definition of a density function, we have $F_j(t|\mathbf{x}) = \int_0^t f_j(s|\mathbf{x})\mathrm{d}s$. Summing over all events yields $\sum_{j=1}^{J} F_j(t|\mathbf{x}) = \int_0^t \sum_{j=1}^{J} f_j(s|\mathbf{x})\mathrm{d}s$. Equation (2.1) implies that $1 - S(t|\mathbf{x}) = \int_0^t \sum_{j=1}^{J} f_j(s|\mathbf{x})\mathrm{d}s$. Taking a derivative with respect to t and Equation (2.3) give that $-\frac{\mathrm{d}}{\mathrm{d}t} S(t|\mathbf{x}) = \sum_{j=1}^{J} f_j(t|\mathbf{x}) = \sum_{j=1}^{J} S(t|\mathbf{x})\alpha_j(t|\mathbf{x})$, which is equivalent to

$\frac{1}{S(t|\mathbf{x})}\frac{\mathrm{d}}{\mathrm{d}t}S(t|\mathbf{x}) = -\sum_{j=1}^{J}\alpha_j(t|\mathbf{x})$. Notice that $\frac{\mathrm{d}}{\mathrm{d}t}\ln S(t|\mathbf{x}) = \frac{1}{S(t|\mathbf{x})}\frac{\mathrm{d}}{\mathrm{d}t}S(t|\mathbf{x})$. Integration from 0 to t gives $\ln S(t|\mathbf{x}) = -\int_0^t \sum_{j=1}^{J}\alpha_j(s|\mathbf{x})\mathrm{d}s$, which is equivalent to

$$S(t|\mathbf{x}) = \exp\left(-\int_0^t \sum_{j=1}^{J}\alpha_j(s|\mathbf{x})\mathrm{d}s\right) = \exp\left(-\sum_{j=1}^{J}\int_0^t \alpha_j(s|\mathbf{x})\mathrm{d}s\right). \tag{2.5}$$

Substitution of Equation (2.5) proves the second equality in Fact (2.4). □

Define the cause-specific cumulative hazard function $H_j(t|\mathbf{x}) := \int_0^t \alpha_j(s|\mathbf{x})\mathrm{d}s$, which gives the accumulated hazard or risk of event j up until time t. Substitution of $H_j(t|\mathbf{x})$ into Equation (2.5) yields

$$S(t|\mathbf{x}) = \exp\left(-\sum_{j=1}^{J} H_j(t|\mathbf{x})\right).$$

2.3. Nelson-Aalen, Kaplan-Meier, and Aalen-Johansen estimators

This chapter introduces non-parametric estimators for the cause-specific cumulative hazard function, the event-free survival function, and the CIF. The estimators are suitable for right-censored competing risks data.

Assume that the covariates $(\mathbf{X}_i)_{1\leqslant i\leqslant n}$ are randomly and independently drawn from a covariate space χ with a joint probability density function $f_{\mathbf{X}}(\mathbf{x})$. Since the estimations do not directly depend on the observed covariates, the estimators provide average estimates for the entire covariate space from which the observed covariates were drawn. Hence, we define

$$S(t) := \mathrm{E}_{\mathbf{X}}S(t|\mathbf{X}) = \mathrm{E}_{\mathbf{X}}P(T^o \geqslant t|\mathbf{X}) = \int_{\chi} P(T^o \geqslant t|\mathbf{x})f_{\mathbf{X}}(\mathbf{x})\mathrm{d}\mathbf{x}, \tag{2.6}$$

$$F_j(t) := \mathrm{E}_{\mathbf{X}}F_j(t|\mathbf{x}) = \mathrm{E}_{\mathbf{X}}P(T^o \leqslant t, \delta^o = j|\mathbf{x}) = \int_{\chi} P(T^o \leqslant t, \delta^o = j|\mathbf{x})f_{\mathbf{X}}(\mathbf{x})\mathrm{d}\mathbf{x}, \tag{2.7}$$

$$\begin{aligned}\alpha_j(t) &:= \mathrm{E}_{\mathbf{X}}\alpha_j(t|\mathbf{X}) = \mathrm{E}_{\mathbf{X}}\lim_{\Delta t\longrightarrow 0+}\frac{P(t\leqslant T^o\leqslant t+\Delta t, \delta^o = j|T^o\geqslant t, \mathbf{X})}{\Delta t}\\ &= \int_{\chi}\lim_{\Delta t\longrightarrow 0+}\frac{P(t\leqslant T^o\leqslant t+\Delta t, \delta^o = j|T^o\geqslant t, \mathbf{x})}{\Delta t}f_{\mathbf{X}}(\mathbf{x})\mathrm{d}\mathbf{x}, \text{and}\end{aligned} \tag{2.8}$$

$$H_j(t) := \mathrm{E}_{\mathbf{X}}H_j(t|\mathbf{X}) = \mathrm{E}_{\mathbf{X}}\int_0^t \alpha_j(s|\mathbf{X})\mathrm{d}s = \int_{\chi}\int_0^t \alpha_j(s|\mathbf{x})\mathrm{d}s f_{\mathbf{X}}(\mathbf{x})\mathrm{d}\mathbf{x}. \tag{2.9}$$

Let $0 \leqslant t_1 < t_2 < ... < t_m$ denote $m \leqslant n$ distinct and ordered realizations of the event times $(T_i)_{1\leqslant i\leqslant n}$. Define $d_j(t_k) = \sum_{i=1}^{n} I(T_i = t_k, \delta_i = j)$, the number of type j events at t_k, $N_j(t) = \sum_{i=1}^{n} I(T_i \leqslant t, \delta_i = j)$, the number of type j events in $[0, t]$,

$d(t_k) = \sum_j d_j(t_k)$, the total number of events occurring at time t_k, $N(t) = \sum_j N_j(t)$, the total number of events occurring in $[0, t]$, and $Y(t) = \sum_{i=1}^{n} I(T_i \geqslant t)$, the number of subjects at risk at time t. A subject i is said to be at risk at time t if neither of the events of interest has happened before time t and the subject has not been censored before time t, i.e., $T_i \geqslant t$.

2.3.1. Nelson-Aalen estimator

The Nelson-Aalen estimator

$$\hat{H}_j(t) = \int_0^t \frac{\mathrm{d}N_j(s)}{Y(s)} = \sum_{k=1}^{m(t)} \frac{d_j(t_k)}{Y(t_k)} \tag{2.10}$$

with $m(t) := \max\{k : t_k \leqslant t\}$ estimates the cumulative cause-specific hazard function $H_j(t)$. The independence of C^o and T^o gives that

$$\begin{aligned}
\alpha_j(t) &:= \lim_{\Delta t \longrightarrow 0+} \frac{1}{\Delta t} P(t \leqslant T^o \leqslant t + \Delta t, \delta^o = j | T^o \geqslant t) \\
&= \lim_{\Delta t \longrightarrow 0+} \frac{1}{\Delta t} \frac{P(t \leqslant T^o \leqslant t + \Delta t, \delta^o = j)}{P(T^o \geqslant t)} \frac{P(C^o \geqslant t + \Delta t)}{P(C^o \geqslant t)} \\
&= \lim_{\Delta t \longrightarrow 0+} \frac{1}{\Delta t} \frac{P(t \leqslant T^o \leqslant t + \Delta t, C^o \geqslant t + \Delta t, \delta^o = j)}{P(T^o \geqslant t, C^o \geqslant t)} \\
&= \lim_{\Delta t \longrightarrow 0+} \frac{1}{\Delta t} P(t \leqslant T^o \leqslant t + \Delta t, C^o \geqslant t + \Delta t, \delta^o = j | T^o \geqslant t, C^o \geqslant t) \\
&= \lim_{\Delta t \longrightarrow 0+} \frac{1}{\Delta t} P(t \leqslant T^o \leqslant t + \Delta t, C^o \geqslant T^o, \delta^o = j | \min(T^o, C^o) \geqslant t) \\
&= \lim_{\Delta t \longrightarrow 0+} \frac{1}{\Delta t} P(t \leqslant T \leqslant t + \Delta t, \delta = j | T \geqslant t).
\end{aligned}$$

This means that $\alpha_j(t)\Delta t$ for small Δt can be estimated as the number of observed subjects experiencing event j in the time interval $[t, t+\Delta t]$ out of the number of subjects at risk at time t. Rewriting this result in the introduced notation yields that

$$\alpha_j(t)\Delta t \approx \frac{N_j(t + \Delta t) - N_j(t)}{Y(t)}. \tag{2.11}$$

Using Equation (2.11) to approximate $H_j(t)$ gives that

$$\begin{aligned}
H_j(t) &:= \mathrm{E}_{\mathbf{X}} \int_0^t \alpha_j(s|\mathbf{X}) \mathrm{d}s \\
&\approx \int_0^t \frac{N_j(s + \mathrm{d}s) - N_j(s)}{Y(s)} \\
&\approx \int_0^t \frac{\mathrm{d}N_j(s)}{Y(s)},
\end{aligned} \tag{2.12}$$

which is an integral with respect to the counting process $N_j(s)$ and where $\mathrm{d}N_j(s)$ denotes the number of observed subjects experiencing event j in the time interval $[s, s + \mathrm{d}s)$. Since integrating over a counting process is the same as summing the integrand over the jump times of the process,

$$\hat{H}_j(t) = \int_0^t \frac{\mathrm{d}N_j(s)}{Y(s)} = \sum_{k=1}^{m(t)} \frac{d_j(t_k)}{Y(t_k)}$$

with $m(t) := \max\{k : t_k \leqslant t\}$.

2.3.2. Kaplan-Meier estimator

The Kaplan-Meier estimator

$$\hat{S}(t) = \prod_{s \leqslant t} \left(1 - \frac{N(\mathrm{d}s)}{Y(s)}\right) = \prod_{k=1}^{m(t)} \left(1 - \frac{d(t_k)}{Y(t_k)}\right) \tag{2.13}$$

with $m(t) := \max\{k : t_k \leqslant t\}$ estimates the event-free survival function. We can write

$$S(t) = P(T^o \geqslant t | T^o \geqslant t_{m(t)}) \cdot \ldots \cdot P(T^o \geqslant t_2 | T^o \geqslant t_1) \cdot P(T^o \geqslant t_1 | T^o \geqslant 0), \tag{2.14}$$

where $P(T^o \geqslant t_1 | T^o \geqslant 0) = P(T^o \geqslant t_1)$ can be estimated by 1 since there is no observation of an event before time t_1. Set $t_{m(t)+1} = t$ for notational convenience and rewrite Equation (2.14) as

$$\begin{aligned} S(t) &\approx \prod_{k=1}^{m(t)} P(T^o \geqslant t_{k+1} | T^o \geqslant t_k) \\ &\approx \prod_{k=1}^{m(t)} \left(1 - P(t_k \leqslant T^o < t_{k+1} | T^o \geqslant t_k)\right) \\ &\approx \prod_{k=1}^{m(t)} \left(1 - \sum_{j=1}^{J} P(t_k \leqslant T^o < t_{k+1}, \delta^o = j | T^o \geqslant t_k)\right). \end{aligned}$$

By Definition (2.8) of the hazard function,

$$\approx \prod_{k=1}^{m(t)} \left(1 - \sum_{j=1}^{J} \alpha_j(t_k) \mathrm{d}t\right)$$

for small $t_k - t_{k-1}$ represented by $\mathrm{d}t$. By Approximation (2.11),

$$\begin{aligned} &\approx \prod_{k=1}^{m(t)} \left(1 - \sum_{j=1}^{J} \frac{\mathrm{d}N_j(t_k)}{Y(t_k)}\right) \\ &\approx \prod_{k=1}^{m(t)} \left(1 - \frac{\mathrm{d}N(t_k)}{Y(t_k)}\right), \end{aligned}$$

where $\mathrm{d}N(t_k)$ denotes the number of observed subjects experiencing any event in the time interval $[t_k, t_k + \mathrm{d}t)$. Hence, define

$$\hat{S}(t) = \prod_{s \leqslant t} \left(1 - \frac{N(\mathrm{d}s)}{Y(s)}\right) = \prod_{k=1}^{m(t)} \left(1 - \frac{d(t_k)}{Y(t_k)}\right)$$

with $m(t) := \max\{k : t_k \leqslant t\}$.

2.3.3. Aalen-Johansen estimator

The Aalen-Johansen estimator

$$\hat{F}_j(t) = \int_0^t \hat{S}(u-)Y^{-1}(u)N_j(\mathrm{d}u) = \sum_{k=1}^{m(t)} \hat{S}(t_{k-1})Y^{-1}(t_k)d_j(t_k) \tag{2.15}$$

with $m(t) := \max\{k : t_k \leqslant t\}$ estimates the cumulative incidence function (CIF) non-parametrically.

Notice that the Kaplan-Meier estimator (2.13) is a step function which allows for left-continuous variant $\hat{S}(u-) = \lim_{t \to u-} \hat{S}(t)$. Substitution of $\hat{S}(u-)$ and Approximation (2.11) to Fact (2.4) gives that

$$F_j(t) \approx \int_0^t \hat{S}(u-)Y^{-1}(u)N_j(\mathrm{d}u).$$

Notice also that $\hat{S}(t_k-) = \hat{S}(t_{k-1})$. Hence,

$$\hat{F}_j(t) = \int_0^t \hat{S}(u-)Y^{-1}(u)N_j(\mathrm{d}u) = \sum_{k=1}^{m(t)} \hat{S}(t_{k-1})Y^{-1}(t_k)d_j(t_k).$$

3. Competing risks forests

A competing risks forest is an ensemble method built on random forests (Breiman, 2001) used for estimation of ensembles of survival data of competing events. Ensemble methods use multiple learning models to obtain better predictive performance, and an ensemble is a resulting estimate from multiple learning models. Specifically, the competing risks forest reduces overfitting by constructing an ensemble from multiple binary decision trees, because a single decision tree tends to overfit on the learning data, and hence, not perform well on unseen data.

3.1. Growth of the competing risks tree

A competing risks tree is a binary decision tree in the setting of competing risks data introduced by Ishwaran et al. (2014). A binary decision tree can be represented by a graph consisting of nodes and branches, where each node represents a splitting rule and each branch represents the resulting split. A node can be either

- an internal node, also called a parent node, which branches into two daughter nodes, or
- a terminal node, also called a leaf node, which does not branch into any daughter nodes.

The goal of the growth of a competing risk tree is to partition given competing risks data into terminal nodes with

- each terminal node containing data with similar covariates and
- data at two different terminal nodes having dissimilar hazard functions.

This partition allows similar data at a terminal node to share the same ensemble estimate of the CIF as described in Construction of ensembles from a competing risks forest.

The growth of the competing risk tree begins at the topmost node, called the root node, with a given competing risks data set. The algorithm recursively splits the data set into two subsets according to a splitting rule until a chosen stop condition is satisfied. This means that for every node, the algorithm either

- stops splitting the data, if the data at the the node satisfy the given stop condition, or
- splits the data at the node into two subsets, each for a new daughter node, according to the given splitting rule.

The competing risk tree growth algorithm uses the stop condition as described by Ishwaran et al. (2014), where $n_0 > 0$ is the minimal number of unique subjects at a node. The stop condition terminates the algorithm if the splitting rule results into a split of the data in which either of the daughter nodes has less than n_0 unique subjects.

3.2. Splitting rule

A splitting rule determines how competing risks data $(T_i, \delta_i, \mathbf{x}_i)_{1 \leqslant i \leqslant n}$ at a parent node are partitioned into two subsets, one for the left daughter node l and the other for the right daughter node r. A good partition maximizes the difference between hazard functions at the left and right daughter nodes. Recursive partitioning leads to a partition of the data at the root node into subsets at terminal nodes with similar hazard functions.

The splitting rule partitions the data with a rule of the form $x \leqslant c$, where x is one of the covariates $x_1, \ldots, x_p$ in the p-dimensional covariate vector $\mathbf{x}$. This means that an observation goes to the left daughter node if and only if its covariate $x \leqslant c$. Otherwise, it goes to the right daughter node. The splitting rule in this section uses notation for continuous covariates, but it also works for categorical covariates. To suppress overfitting, the competing risks forest algorithm randomly selects $M \leqslant p$ candidate covariates before splitting at each node. Only a candidate covariate can be used in the splitting rule $x \leqslant c$.

3.2.1. Generalized log-rank test

The generalized log-rank test selects a candidate covariate x and a value c that maximize a weighted difference of estimates of a cause-specific cumulative hazard function of each daughter node determined by a rule of the form $x \leqslant c$. Its splitting rule $L_j^{LR}(x, c)$ tests the null hypothesis

$$H_0 : \alpha_{jl}(t) = \alpha_{jr}(t)$$

for all $t \leqslant t_m$, where $\alpha_{jl}(t)$ and $\alpha_{jr}(t)$ are the the cause-j specific hazard functions in the left and the right daughter nodes, respectively.

Define x_i as the value of a candidate covariate x for subject i; $Y_l(t) = \sum_{i=1}^n I(T_i \geqslant t, x_i \leqslant c)$ and $Y_r(t) = \sum_{i=1}^n I(T_i \geqslant t, x_i > c)$ as the number of subjects at risk at time t in the left and right daughter node, respectively; $Y(t) = Y_l(t) + Y_r(t)$ as the number of subjects at risk at time t at the parent node; $d_{j,l}(t) = \sum_{i=1}^n I(T_i = t, \delta_i = j, x_i \leqslant c)$ and $d_{j,r}(t) = \sum_{i=1}^n I(T_i = t, \delta_i = j, x_i > c)$ as the number of type j events at t in the left and right daughter node, respectively; and $d_j(t) = d_{j,l}(t) + d_{j,r}(t)$ as the number of type j events at time t at the parent node.

The splitting rule is given by

$$L_j^{LR}(x, c) = \frac{1}{\hat{\sigma}_j^{LR}(x, c)} \sum_{k=1}^{m} W_j(t_k) \left(d_{j,l}(t_k) - \frac{d_j(t_k) Y_l(t_k)}{Y(t_k)} \right) \tag{3.1}$$

where time-dependent weights $W_j(t) > 0$ are used to make the test more sensitive to early or late differences between the cause-specific hazard functions, and

$$\left(\hat{\sigma}_j^{LR}(x,c)\right)^2 = \sum_{k=1}^{m} W_j(t_k)^2 d_j(t_k) \frac{Y_l(t_k)}{Y(t_k)} \left(1 - \frac{Y_l(t_k)}{Y(t_k)}\right) \left(\frac{Y(t_k) - d_j(t_k)}{Y(t_k) - 1}\right).$$

is the variance estimate. Notice that

$$\begin{aligned}
L_j^{LR}(x,c) &= \frac{1}{\hat{\sigma}_j^{LR}(x,c)} \sum_{k=1}^{m} W_j(t_k) \left(d_{j,l}(t_k) - \frac{d_j(t_k) Y_l(t_k)}{Y(t_k)}\right) \\
&= \frac{1}{\hat{\sigma}_j^{LR}(x,c)} \sum_{k=1}^{m} W_j(t_k) \left(\frac{d_{j,l}(t_k)(Y_l(t_k) + Y_r(t_k)) - (d_{j,l}(t_k) + d_{j,r}(t_k)) Y_l(t_k)}{Y_l(t_k) + Y_r(t_k)}\right) \\
&= \frac{1}{\hat{\sigma}_j^{LR}(x,c)} \sum_{k=1}^{m} W_j(t_k) \left(\frac{d_{j,l}(t_k) Y_r(t_k)}{Y(t_k)} - \frac{d_{j,r}(t_k) Y_l(t_k)}{Y(t_k)}\right) \\
&= \frac{1}{\hat{\sigma}_j^{LR}(x,c)} \sum_{k=1}^{m} W_j(t_k) \frac{Y_r(t_k) Y_l(t_k)}{Y(t_k)} \left(\frac{d_{j,l}(t_k)}{Y_l(t_k)} - \frac{d_{j,r}(t_k)}{Y_r(t_k)}\right),
\end{aligned}$$

is a summation of weighted differences of Nelson-Aalen estimators (2.10) for each daughter node. Therefore, maximizing $|L_j^{LR}(x,c)|$ over x and c provides a partition of the data into two subsets with dissimilar cause-j hazard functions.

3.2.2. Composite splitting rules

The following composite splitting rule is cause-nonspecific because it selects a candidate covariate x and a value c by considering all cause-specific $L_j^{LR}(x,c)$. This grows a tree whose each terminal node contains subjects with similar hazard functions of all causes. In particular, this thesis uses the splitting rule given by maximization of $|L^{LR}(x,c)|$ over x and c where

$$L^{LR}(x,c) = \frac{\sum_{j=1}^{J} \left(\hat{\sigma}_j^{LR}(x,c)\right)^2 L_j^{LR}(x,c)}{\sqrt{\sum_{j=1}^{J} \left(\hat{\sigma}_j^{LR}(x,c)\right)^2}}. \tag{3.2}$$

3.3. Construction of ensembles from a competing risks forest

The competing risks forest is a collection of B grown competing risks trees. The randomization is introduced by growing each competing risks tree using an independent bootstrap sample $b \in \{1, \cdots, B\}$ of the learning data $(T_i, \delta_i, \mathbf{x}_i)_{1 \leqslant i \leqslant n}$. Each bootstrap sample is a random selection of n subjects with replacement from the learning data. Sampling with replacement results in exclusion of on average 37% of the original competing risks data. The excluded data is called the out-of-bag data (OOB data).

Suppose we want to estimate CIF $F_j(t|\mathbf{x})$ and survival function $S(t|\mathbf{x})$ for a subject with covariates $\mathbf{x}$. Drop the covariate $\mathbf{x}$ down a tree grown from, say, bootstrap sample b. Then $\mathbf{x}$ follows a splitting rule at each parent node of tree b until it ends up in a unique terminal node. Let $h_b(\mathbf{x})$ be the set of indices of observations from the learning data whose covariates share the terminal node with $\mathbf{x}$ and $c_{i,b}$ be the number of times observation i occurs in bootstrap sample b. Then $\mathbf{x}$'s estimate of the event-free survival function for tree b is given by Kaplan-Meier estimator (2.13)

$$\hat{S}_b(t|\mathbf{x}) = \prod_{s \leqslant t} \left(1 - \sum_{j=1}^{J} \frac{N_{j,b}(\mathrm{d}s|\mathbf{x})}{Y_b(s|\mathbf{x})} \right),$$

where $N_{j,b}(t|\mathbf{x}) = \sum_{i \in h_b(\mathbf{x})} c_{i,b} I\{T_i \leqslant t, \delta_i = j\}$ and $Y_b(t|\mathbf{x}) = \sum_{i \in h_b(\mathbf{x})} c_{i,b} I\{T_i \geqslant t\}$, and $\mathbf{x}$'s estimate of the CIF for tree b is given by Aalen-Johansen estimator (2.15)

$$\hat{F}_{j,b}(t|\mathbf{x}) = \int_0^t \hat{S}_b(u - |\mathbf{x}) Y_b^{-1}(u|\mathbf{x}) N_{j,b}(\mathrm{d}u|\mathbf{x}).$$

The ensemble estimates for a subject with covariates $\mathbf{x}$ is an average of the estimates for each tree in the forest, i.e.,

$$\bar{S}(t|\mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} \hat{S}_b(t|\mathbf{x}) \text{ and } \bar{F}_j(t|\mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} \hat{F}_{j,b}(t|\mathbf{x}).$$

3.4. Comparison to the COX model

The cause-specific COX model (Cox, 1975) a semiparametric statistical model that relates the cause-specific hazard function (2.2) to a covariate vector $\mathbf{x}$ with equation

$$\alpha_j(t|\mathbf{x}) = \alpha_{j,0}(t) \exp\left(\beta_j^T \mathbf{x}\right) \tag{3.3}$$

where $\alpha_{j,0}(t)$ is the nonparametric part called a baseline hazard and $\exp\left(\beta_j^T \mathbf{x}\right) = \exp\left(\beta_{j,1} x_1 + ... + \beta_{j,p} x_p\right)$ is the parametric part called a hazard ratio. Each parameter $\beta_{j,l}$ describes how the corresponding covariate x_l affects the hazard function $\alpha_j(t|\mathbf{x})$. Estimation of parameters β_j for every cause j allows for consequent estimation of the CIF and survival function with Equations (2.4) and (2.5).

Observe that two subjects with covariate vectors $\mathbf{x}_1$ and $\mathbf{x}_2$ must then have proportional hazard functions because

$$\frac{\alpha_j(t|\mathbf{x}_1)}{\alpha_j(t|\mathbf{x}_2)} = \exp\left(\beta_j^T (\mathbf{x}_1 - \mathbf{x}_2)\right)$$

is time-constant. This is a rather unrealistic consequence of the assumptions in the COX model. Recall that competing risks forests are a nonparametric method without strict assumptions on the form of the hazard functions. Therefore, competing risks forests are advantageous in this regard.

Another point of comparison is interpretability. The COX model shows how much effect each covariate has on the hazard function by the corresponding paramater. On the other hand, competing risks forests act as a "black box" because of complex internal workings of the model without easy and clear interpretation compared to the COX model. Hence, the COX model is more advantageous from the interpretability point of view.

3.5. Application to medical data

This section uses competing risks forests and the cause-specific COX model (3.3) to analyze follicular cell lymphoma data (Pintilie, 2006). Analyses were performed in R programming language version 4.3.0 (R Core Team, 2023) using R-packages `randomForestSRC` (Ishwaran and Kogalur, 2023) and `survival` (Therneau, 2023). The code is available in Appendix R-Code.

Follicular cell lymphoma is a cancer that originates from the uncontrolled division of certain types of B-cells. The 541 patients in the dataset (Pintilie, 2006) were treated with only radiotherapy or both radiotherapy and chemotherapy. Each patient eventually either

- relapses, which means that the patient dies during the treatment or relapses after the treatment, or
- dies, which means that the patient is successfully treated and dies for other reasons than follicular cell lymphoma.

These two events form two competing risks for each patient. The time when either of the events occurs or a patient is censored is recorded. The dataset also contains an age, a hemoglobin level (g/l), and a clinical stage (I or II) of the cancer of each patient at the time of diagnosis.

We grow a competing risks forest with 100 trees using the composite splitting rule (3.2). Figure 3.1 shows one of the competing risks trees as a graph. Suppose we are interested in the effect of chemotherapy on relapse depending on the diagnosed stage of cancer of a patient. The competing risks forest algorithm estimates the CIFs for relapse given the treatment and clinical stage of cancer as shown in Figure 3.2. Figure 3.2 suggests that a patient who underwent both chemotherapy and radiotherapy is more likely to have relapsed in the first year after the start of the treatment compared to a patient who only underwent radiotherapy. After the first year, however, a patient who underwent both treatments is less likely to have relapsed compared to a patient who underwent only radiotherapy. In summary, the inclusion of chemotherapy to the treatment of cancer by radiotherapy seems to decrease the chance of a relapse after the first year, but seems to increase the chance of a relapse in the first year, for patients with either clinical stage of cancer.

For comparison, fit the cause-specific COX model (3.3) with covariates age, hemoglobin level, clinical stage, and chemotherapy. The COX model yields estimates of CIFs for

relapse given the stage of cancer and patient's treatment with chemotherapy. Figure 3.3 shows the CIF estimates for an average subject in the follicular cell lymphoma dataset. The average subject is 57 years old and has hemoglobin levels of 139 g/l. Figure 3.3 suggests that patients who undergo chemotherapy have a lower chance to have experienced a relapse compared to patients who do not undergo chemotherapy. This seems to hold in both the short and long run, and both stages of cancer. That is a difference to the competing risks forest model, because the competing risks forest model suggests that the chance to have relapsed is elevated in the first year when patients undergo chemotherapy. Another difference is that all CIFs in Figure 3.2 are higher than the corresponding CIFs in Figure 3.3.

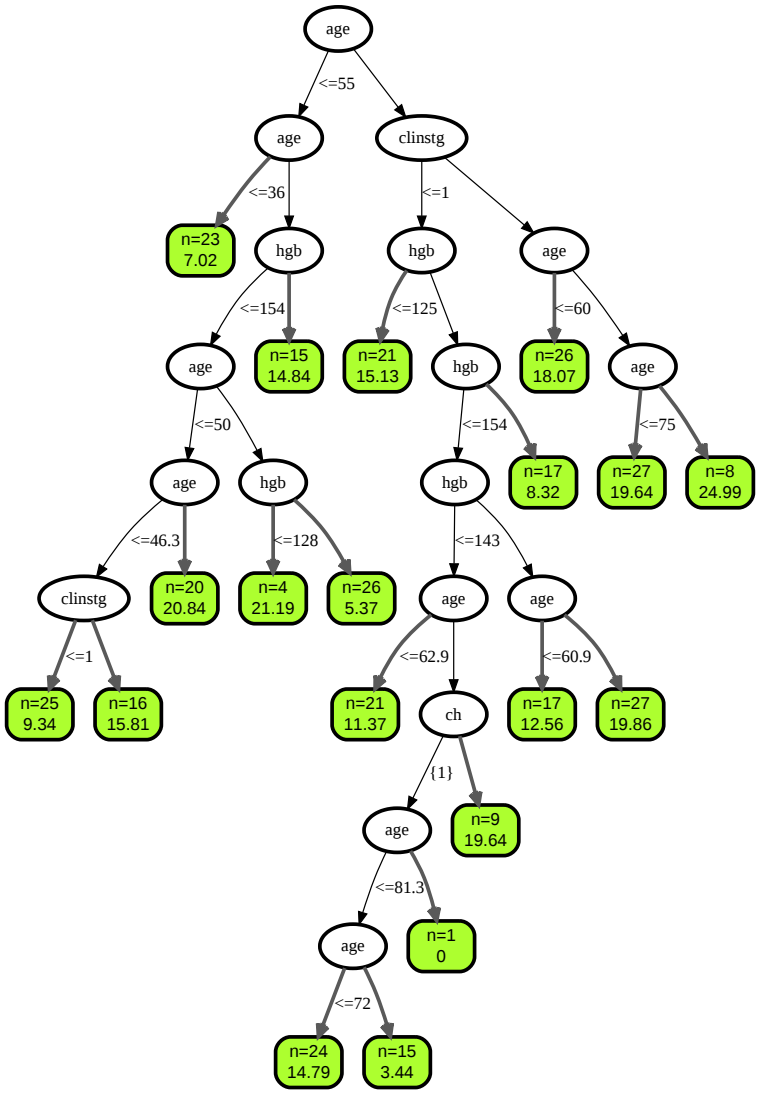

Figure 3.1.: A competing risks tree with green terminal nodes, a splitting covariate x in each parent node, and a splitting value c between each pair of branches.

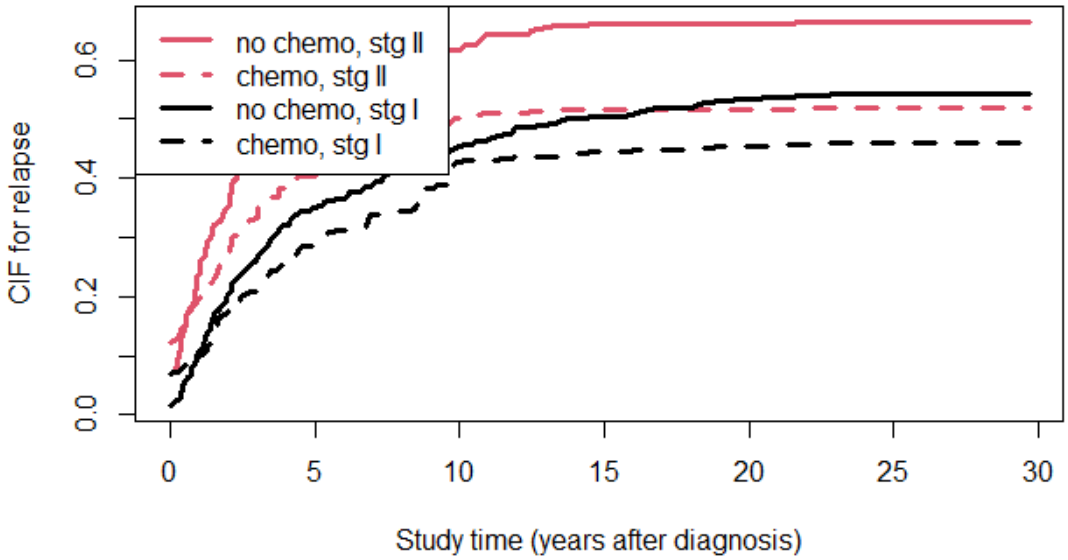

Figure 3.2.: A plot of ensemble estimates of CIFs for relapse in four scenarios: 1. Patients diagnosed with stage II cancer undergo only radiotherapy. 2. Patients diagnosed with stage II cancer undergo the combined treatment by chemotherapy and radiotherapy. 3. Patients diagnosed with stage I cancer undergo only radiotherapy. 4. Patients diagnosed with stage I cancer undergo the combined treatment by chemotherapy and radiotherapy.

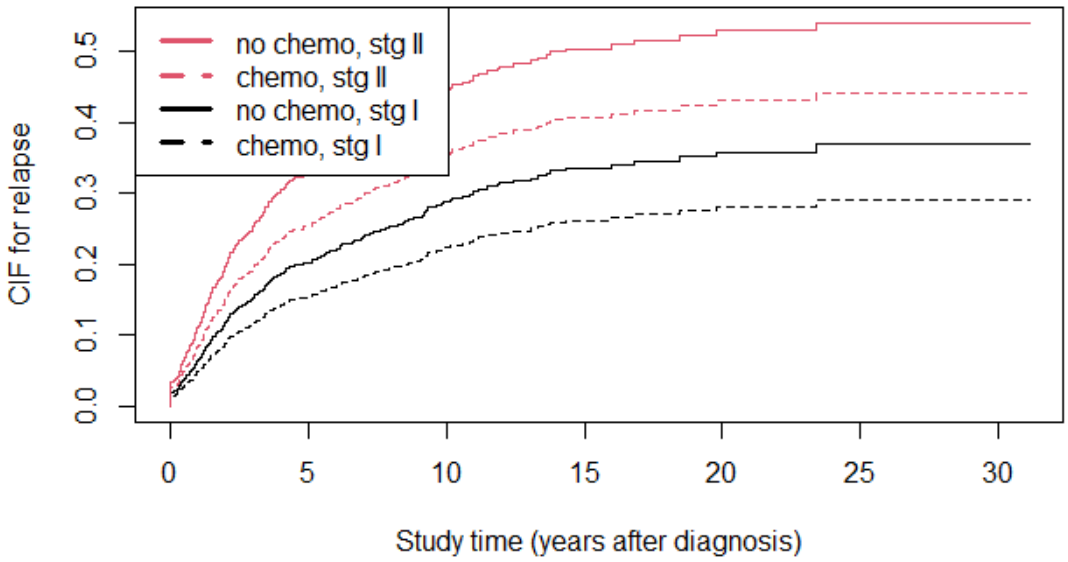

Figure 3.3.: A plot of estimates of CIFs for relapse from the COX model in the four scenarios mentioned in Figure 3.2.

A. R-Code

```
# Competing risks forest
library(randomForestSRC)
data(follic , package = "randomForestSRC")

follic.obj <- rfsrc(Surv(time, status) ~ .,
                    follic ,
                    nsplit = 3,
                    ntree = 100)

cif <- follic.obj$cif.oob
Time <- follic.obj$time.interest
ch <- follic$ch
clinstg <- follic$clinstg


cif.rel <- cbind(
            apply(cif[,,1][ch == "N" & clinstg == 2,], 2, mean),
            apply(cif[,,1][ch == "Y" & clinstg == 2,], 2, mean),
            apply(cif[,,1][ch == "N" & clinstg == 1,], 2, mean),
            apply(cif[,,1][ch == "Y" & clinstg == 1,], 2, mean))

matplot(Time, cif.rel,
        type = "l",
        lty = c(1,2,1,2),
        col = c(2,2,1,1),
        lwd = 3,
        ylab = "CIF for relapse",
        xlab = "Study time (years after diagnosis)")

legend("topleft",
       legend = c("no chemo,
                  stg II", "chemo, stg II",
                  "no chemo, stg I", "chemo, stg I"),
       lty = c(1,2,1,2),
       col = c(2,2,1,1),
       lwd = 3,
```

```
        cex = 1)

# cause-specific COX model
library(survival)

data = cbind(id = 1:nrow(follic), follic)

data$status <- as.factor(data$status)

cox_model <- coxph(Surv(time, status) ~ age + hgb + clinstg + ch,
                   data,
                   id=id)

plot(survfit(cox_model, newdata = data))

dummy <- expand.grid(ch=c("N", "Y"),
                   age = 57,
                   hgb = 139,
                   clinstg = c("1", "2"))

relapse_cif <- survfit(cox_model, newdata = dummy)[,2]

plot(relapse_cif,
    lty = c(1,2,1,2),
    col = c(1,1,2,2),
    xlab = "Study time (years after diagnosis)",
    ylab = "CIF for relapse")

legend("topleft",
        legend = c("no chemo, stg II",
        "chemo, stg II",
        "no chemo, stg I",
        "chemo, stg I"),
        lty = c(1,2,1,2),
        col = c(2,2,1,1),
        lwd = 3,
        cex = 1)
```

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